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DPR Lecture: Neurologic Exam 3: Sensory System, Special Techniques and Documentation

Richard Morgan, D.O.
DPR 1 Coordinator
Department of Family Medicine
rmorga04@nyit.edu

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Session Objectives

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- Learn how to perform a neurological assessment of the Sensory System and the use of Special Techniques
- Learn how to obtain a detailed health history in a patient with a neurological complaint
- Review how to document the physical findings for a neurological examination and include the sensory system

Source: Course Syllabus

Components of a Screening Neurological Examination

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- **Mental Status**—level of alertness, appropriateness of response, orientation to date and place
- **Cranial Nerves**
- Visual Acuity
- Pupillary light reflex
- Eye movements
- Hearing
- Facial strength- smile, eye closure
- Speech
- **Motor System**
- Strength—shoulder abduction, elbow flexion/extension, wrist extension, finger abduction, hip flexion, knee flexion/extension, ankle dorsiflexion
- **Gait**—casual and tandem
- **Coordination**—fine finger movements, finger-to-nose or finger to chin
- **Reflexes-** Deep Tendon Reflexes (biceps, patellar and achilles) and Plantar responses
- **Sensory System**—one modality at toes—can be light touch, pain/temperature, or proprioception

Bates, 13th Edition, Chapter 24- American Academy of Neurology Guidelines for a Screening Neurologic Examination

Sensory Pathways

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Sensory impulses:

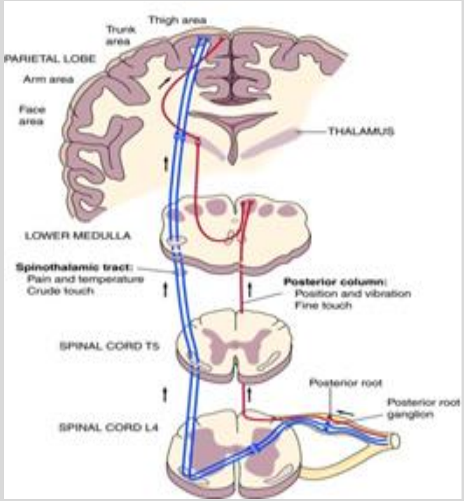
Participate in reflex activity, conscious sensation, locate body position in space, help regulate internal autonomic functions: heart rate, respiration and blood pressure

Reach the Sensory Cortex of the brain via

- Spinothalamic tract (smaller sensory neurons that are unmyelinated or thinly myelinated axons)- pain, temperature crude touch
- Posterior columns (larger neurons with heavily myelinated axons)- vibration, proprioception, kinesthesia, pressure, fine touch (skin and joint position receptors)

*Both the Spinothalamic tract and Posterior columns will have a cross-over to the opposite side as it travels to the Thalamus.

See diagram on slide- Spinothalamic tract (blue) and Posterior column (red)



Sensation

- Analgesia- absence of pain sensation
- Hypalgesia (or Hypoalgesia)- decreased sensitivity to pain
- Hyperalgesia- increased pain sensitivity
- Anesthesia- absence of touch sensation
- Hypesthesia- decreased sensitivity to touch
- Hyperesthesia- increased sensitivity to touch
- Paresthesia- sensation of tingling or “pins and needles”

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Sensory System

Sensations that are tested:

- Pain and Temperature (Spinothalamic tracts)
- Position (proprioception) and Vibration (Posterior columns)
- Light touch (Spinothalamic and Posterior columns)
- Discriminative sensations -depends on some of the sensations listed above and also processing by the Sensory Cortex

Sensory Testing

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- Comparison of sides- symmetric areas
- Comparison of distal vs proximal of the extremities- pain, temperature and touch
- Vibration and position sensation- test fingers and toes (distal) first
- Vary pace of testing
- Map out the boundaries of sensory loss or hypersensitivity

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Dermatomes and Peripheral Nerve Patterns

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- Dermatomes- A band of skin innervated by the sensory root of a single spinal nerve
- Peripheral Nerve patterns- indicate areas innervated by a specific nerve
- Knowledge of the dermatomes and peripheral nerve patterns help to localize the neurologic lesion to a specific level of the spinal cord or nerve that is affected

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Sensory

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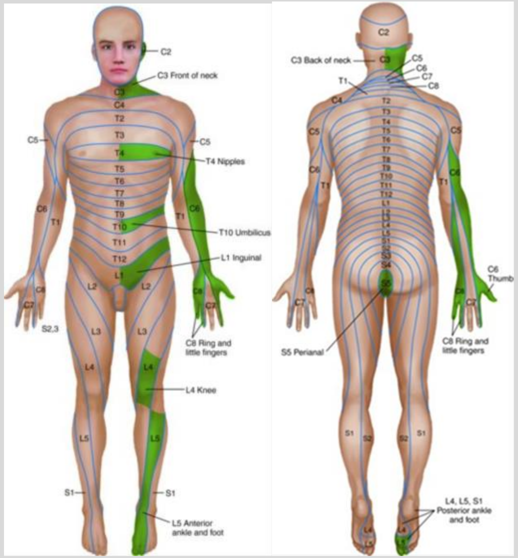
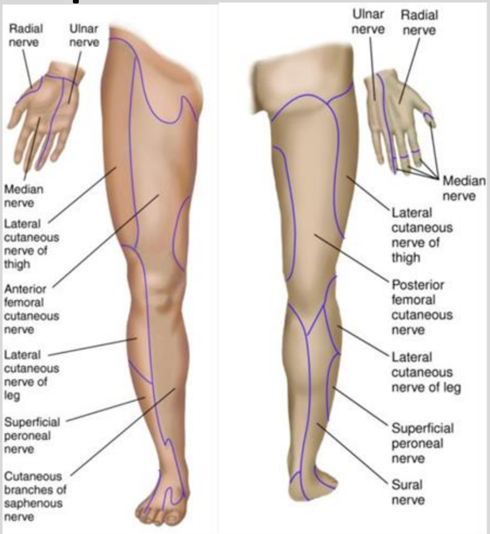
Dermatomes



Dermatomes innervated by

Posterior roots

Peripheral Nerve Pattern



Areas innervated by

Peripheral nerves

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Sensory Testing

- Pain
- Temperature- hot or cold
  (only tested if pain sensation is abnormal)
- Light touch
- Vibration
- Proprioception
- Discriminative Sensations

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Sensory Testing- Dermatomes

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Pain

- Touch multiple areas of the body with a sharp/semi-sharp object or soft object and have the patient identify what it is and its location with their eyes closed

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Sensory Testing

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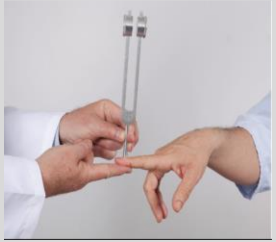
Vibration

- Use a low pitched vibrating tuning fork (128 Hz)
- Place tuning fork over the distal interphalangeal joint (IP)- finger or toe
- If vibration is impaired test a more proximal bony prominence ex: wrist, elbow, medial malleolus, shin, patella, anterior superior iliac spine, spinous processes and clavicles

*Often the first sensation lost in peripheral neuropathy



Proprioception

Joint position sense

- Test fingers and toes first
- If distal joints are normal, most likely proprioception is intact in proximal joints

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Discriminative Sensations

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- Tests the ability of the sensory cortex to correlate, analyze and interpret sensations
- Touch and position sense need to be intact for tests
- Absence or a decrease in discriminative sensation indicates a lesion in the sensory cortex
- Patient’s eyes should be closed for all tests

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Discriminative Sensations

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Stereognosis

- Identify object through touch

Graphesthesia (number identification)

- Identify the number drawn on patient's palm

Two Point Discrimination

- Touch two places simultaneously using open ends of paper clip
- Normal minimal distance for discrimination varies for different parts of the body

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Swartz, 8th Edition Chapter 21

Discriminative Sensations

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Point Localization

- Patient has eyes closed
- Briefly touch a point on the patient’s skin
- Ask the patient to open their eyes and point to the place touched
- If the patient has a lesion of the sensory cortex, their ability to localize the point(s) is impaired

Extinction

- Patient has eyes closed
- Simultaneously stimulate areas on the both sides of the body.
- Patient will identify the location of the stimuli

With Sensory Cortex lesions, only one stimulus may be felt. The stimulus that is not recognized, is “extinguished”, and is on the contralateral side of the affected sensory cortex.

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Swartz 8th Edition Chapter 21

Special Techniques

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- Brudzinski Sign
- Kernig Sign
- Axial Compression Test
- Foraminal Distraction Test
- Straight Leg Raise (with Bragard's Test)
- Spurling Test
- Well (Opposite) Straight Leg Raising Test
- Hoover Test

Meningeal Signs

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- Test when suspect meningeal inflammation from either meningitis or subarachnoid hemorrhage
- Brudzinski Sign
- Kernig Sign

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Brudzinski and Kernig Sign

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Brudzinski Sign

- Patient's head and neck are flexed forward
- Normal response- Hips and knees should remain relaxed
- Positive Brudzinski Sign- Flexion of both hips and knees

Kernig Sign

- Patient's leg is flexed at the hip and knee
- Slowly extend the leg and straighten the knee
- Positive Kernig's Sign- Pain and increased resistance to knee extension

Axial Compression Test

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Test for cervical neck impingement

- Patient is seated
- Examiner places both hands on top of the patient’s head
- Exert a downward force with both hands

*Test is positive if pain/neurological symptom are reproduced in the cervical spine and/or upper extremity

Hoppenfeld Chapter 4

Foraminal Distraction Test

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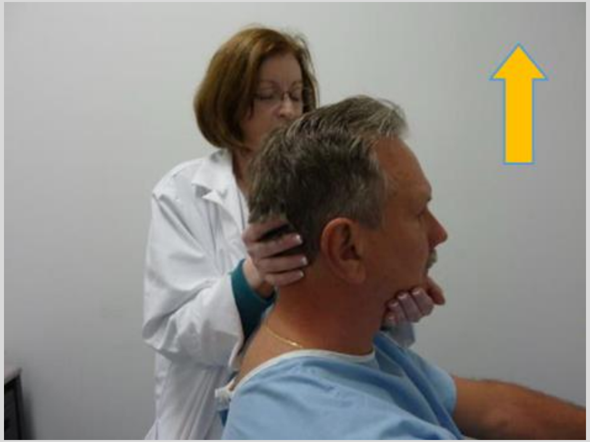
Test for cervical nerve impingement

- Patient is seated
- Examiner places one hand under the patient’s chin and the other hand under the occiput
- An upward traction force is applied on the cervical spine



*Test is positive if the pain/neurological symptoms of the cervical spine and/or upper extremity are reduced with the traction force

Hoppenfeld Chapter 4

Spurling Test

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Test for Cervical Nerve Root Compression

- Patient looks over their shoulder and then up at the ceiling
- Downward pressure is applied to the patient’s head
- Maneuver reproduces the neck pain with radiation to the arm of the same side that the head is turned to If there is pain with just the positioning of the head It is not necessary to apply downward pressure of the head

Bates Chapter 23

Straight Leg Raise (with Bragard’s Test)

Tests for radicular sciatic nerve pain

- Patient is supine with legs extended
- Test unaffected side first
- Raise the patient’s relaxed and straightened leg by the ankle and place a hand on the knee to prevent it from bending.
- Stop when the patient feels pain or maximal flexion has been achieved. (60-120 degrees)
- Repeat on affected side.

Bragard’s Test- lower the leg to the point just below the level of pain and add dorsiflexion of the foot

L5 portion of the sciatic nerve will be stretched and cause pain that radiates below the knee

DeJong 8th Edition Chapter 47

NYITCOM Neurological Physical Exam Skills Sheet

Well (Opposite) Straight Leg Raising (Crossed Straight Leg Raise)

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Test differentiates between sciatic nerve irritation and a herniated disc space occupying lesion

- Patient is supine with legs extended
- Examiner asks the patient to raise their fully extended leg on the uninvolved

* The test may indicate a possible space occupying lesion if the pain occurs on the opposite side of the leg that was raised

DeJong 8th Edition Chapter 47

Hoover Test

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Test for malingering

- Patient is supine
- Unilateral exam
- Examiner places their hands under the patient’s calcaneus bilaterally (see photo)
- Patient is instructed to raise the leg of the involved side without bending their knee
- Examiner is monitoring for a downward pressure of the heel on the side not being tested

*Test is positive if there is an absence of a downward pressure of the uninvolved foot when the patient is raising the involved leg

Hoppenfeld Chapter 9

Neurological Exam

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Start with the patient’s history

Common or Concerning Symptoms:

- Headache
- Dizziness/Vertigo
- Weakness
- Numbness, Abnormal or Absent Sensation
- Syncope/ Near Syncope
- Seizures
- Tremors/ Involuntary Movements

Common Neurological Complaints

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- **Headache**
  - OPQRSTA
    - Onset, History of head trauma
    - Severity *very important to assess "worst headache of my life"
  - Location
  - Duration
  - Aura or Prodrome
  - Associated symptoms
    - Visual changes-double vision
    - Photo or Phonophobia
    - Weakness
    - Loss of sensation
    - Worse with coughing, sneezing, head movement (alteration of intracranial pressure)
    - Infectious symptoms: fever, stiff neck, sinus congestion, sore throat, ear pain
  - Take a thorough past history also including medications (headache could be a side effect or a withdrawal symptom)
  - Lifestyle changes: increased/decreased caffeine, nicotine, alcohol, stress
  - Recommend patient keep a headache diary to note triggers

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Common Neurological Complaints

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- **Dizziness and Vertigo**
  - OPQRSTA
    - Important to distinguish between feeling lightheaded or weak in the legs vs a spinning sensation
  - Age of the patient
  - Onset
  - Duration
  - Associated symptoms
    - Sensation that the room is spinning
    - Nausea/ Vomiting
    - Problems hearing/ ringing in the ears
    - Double vision
    - Problem with balance or gait
  - Take a thorough history also including medications (can be a side effect)

Bates, 13th Edition Chapters 13 and 24

Common Neurological Complaints

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- **Weakness**
  - OPQRSTA
    - Onset *very important to assess
      - Abrupt onset of motor or sensory deficit occurs in TIA and CVA
  - Location
    - Area of the body
    - Generalized or localized (face, limbs)
    - Unilateral or bilateral
    - Look for patterns of weakness
      - Proximal muscles (shoulder, hip girdle)
      - Distal muscles (hands and feet)
      - Symmetrical (same area of body bilaterally)
      - Asymmetrical
  - Associated symptoms
    - Fatigue
    - Apathy
    - Drowsiness
    - Loss of strength
    - Mental status changes
    - Slurred speech
  - Take a thorough past history

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Common Neurological Complaints

- **Numbness, Abnormal or Absent sensation**
  - OPQRSTA
  - Quality- Describe the numbness
      - Is it tingling like pins and needles? (parasthesias)
      - Distorted sensation? (dysesthesias)
      - Reduced or completely absent sensation? (anesthesia)
    - Burning
    - Irritating sensation
  - Pattern
    - Stocking-glove distribution
    - Dermatomal
    - Patchy areas of sensory loss in limbs
  - Take a thorough past history
    - Injury, accident, MVA, trauma
    - Medical history- DM, nerve compression or entrapment

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Common Neurological Complaints

- **Near-syncope and Syncope**
    - OPQRSTA
    - Important to assess "Was there loss of consciousness?"
    - Near-Syncope
        - Hear external noises or voices throughout episode
        - Lightheaded
        - Weakness
    - Syncope
        - Complete loss of consciousness
    - What were you doing at time of onset?
    - Standing, sitting or lying down?
    - Any triggers or warning symptoms?
    - How long did the episode last?
    - How quickly did the onset occur?
    - Was the episode witnessed?
    - Associated symptoms
        - Palpitations- cardiac arrhythmia
        - Diaphoretic
        - Nausea
        - Pallor
        - Loss of bowel/bladder control
    - Take a thorough past history including PMH: heart disease, seizures and medications-

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Common Neurological Complaints

- **Seizures**
  - OPQRSTA
  - Important to assess
    - Onset
      - Occurred once or multiple times
      - Loss of consciousness
      - Unilateral or Bilateral
      - Part of the body affected
      - Age of patient
      - Head trauma
      - Witnessed
  - Associated Symptoms
    - Bladder or bowel incontinence
    - Drowsiness or impaired memory after the event
    - Tongue biting
    - Fever
  - Medical /Social History
    - h/o Head Trauma, Stroke, DM, Meningitis, Encephalitis ,Uremia
    - Alcohol, Cocaine, Benzodiazepine, Barbiturate use/ withdrawal

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Common Neurological Complaints

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- **Tremors and Involuntary Movements**
  - OPQRSTA
    - Tremors or Involuntary movements- Oral- Facial Dyskinesias, Tics, Dystonia, Athetosis, Chorea,
  - Location
    - Area of the body
  - Characteristics
    - Occurs at rest, maintaining a posture or with intentional movement
    - Repetitive
    - Coordinated
    - Increase or decrease with voluntary movement
    - Type of movement
  - Medical/Social History/Family History
    - Medical history
    - Medications
    - Family members

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http://arktos.nyit.edu/login?url=https://meded.lwwhealthlibrary.com/content.aspx?sectionid=248762241&bookid=2964#248762926
Table 24.8

Tics and Involuntary Movements

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Oral–Facial Dyskinesias
Oral–facial dyskinesias are arrhythmic, repetitive, bizarre movements that chiefly involve the face, mouth, jaw, and tongue: grimacing, pursing of the lips, protrusions of the tongue, opening and closing of the mouth, and deviations of the jaw. The limbs and trunk are involved less often. These movements may be a late complication of antipsychotic or antiemetic drugs such as phenothiazines, termed *tardive* (late) dyskinesias. They also occur in long-standing psychoses, in some older adults, and in some edentulous persons.

Tics
Tics are brief, repetitive, stereotyped, coordinated movements occurring at irregular intervals. Examples include repetitive winking, grimacing, and shoulder shrugging. Causes include Tourette syndrome and late effects of drugs such as phenothiazines.

Dystonia
Dystonia causes irregular movements resembling athetosis or tremor. These are often accompanied by abnormal postures that limit voluntary movement and can at times be painful. Examples include writer's cramp, blephorospasm, and as illustrated, spasmodic torticollis.

Athetosis
Athetoid movements are slower and more twisting and writhing than choreiform movements and have a larger amplitude. They most commonly involve the face and the distal extremities. Athetosis is often associated with spasticity. Causes include cerebral palsy.

Chorea
Choreiform movements are brief, rapid, jerky, irregular, and unpredictable. They occur at rest or interrupt normal coordinated movements. Unlike tics, they seldom repeat themselves. The face, head, lower arms, and hands are often involved. Causes include Sydenham chorea (with rheumatic fever) and Huntington disease.

Bates 13th edition Chapter 24 Table 24-8

Preventable Measures- Ischemic Stroke

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Major modifiable risk factors increasing risk for Ischemic Stroke:

- Hypertension
- Diabetes mellitus
- Smoking
- Dyslipidemia
- Physical inactivity

http://arktos.nyit.edu/login?url=https://www.uptodate.com/contents/overview-of-secondary-prevention-of-ischemic-stroke?search=cva%20prevention&source=search_result&selectedTitle=2~150&usage_type=default&display_rank=2

Screening for Diabetic Peripheral Neuropathy

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- Diabetic patients are unaware of neuropathy
- Unaware of sensory loss
- Distal symmetric polyneuropathy is the most common neuropathy for these patients

  Slowly progressive and often asymptomatic

  Risk factor for ulceration, arthopathy and amputation

  Prevalence- 20% in long term type I diabetic patients

  50% in long term type II diabetic patients
- Important to maintain tight glycemic control, routine foot examinations assessing for neuropathy (including- Temperature or Pinprick sensation, Proprioception, ankle reflexes, vibration perception (128 Hz tuning fork) and plantar light touch sensation using a 10 –G monofilament), skin breakdown, poor circulation and musculoskeletal abnormalities

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Documentation of a Normal Neurological Examination

Neuro-

Mental Status: The patient is alert, attentive, and oriented to person, place and time. Speech; clear and fluent with normal comprehension

- CN I not tested CN II-XII grossly intact
- Motor strength is 5/5 in all extremities, good muscle bulk and tone
- Rapid alternating movements and finger to nose are intact
- Gait is normal, no evidence of ataxia
- Negative Romberg and Pronator Drift
- Sensory- Pinprick, light touch, position and vibration intact
- Deep Tendon Reflexes 2+ bilaterally, plantar reflexes downgoing or plantar flexion bilaterally

Bates, 13th Edition Chapter 24
Swartz, 7th Edition, Chapter 19

Documentation of a Normal Cranial Nerve Examination

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Neurologic Examination

- Mental Status: The patient is alert, attentive, and oriented to person, place and time. Speech; clear and fluent with normal comprehension
- Cranial Nerves:

CN I- Not tested

CN II- Visual fields are full to confrontation. Fundoscopic exam is normal with sharp discs and no vascular changes. Visual acuity is 20/20 bilaterally

CN III, IV, VI- Pupils 4 mm and reactive to light; extraocular muscles intact (EOMI), no ptosis

https://neuro.wustl.edu/Education/Medical-Student-Education/Neurology-Clerkship/History/Sample-Write-Ups
Bates, 12th Edition Chapter 17

Documentation of Cranial Nerve Examination- (continue)

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CN V- Facial sensation is intact to pinprick in all 3 divisions bilaterally, Corneal reflexes are present

Temporal and Masseter strength intact

CN VII- Face is symmetric with normal eye closure and smile

CN VIII- Hearing intact bilaterally to rubbing fingers (or whispered voice)

CN IX, X- Palate elevates symmetrically, phonation is normal, gag reflex intact

CN XI- Head turning and shoulder shrug intact

CN XII- Tongue midline, no atrophy

https://neuro.wustl.edu/Education/Medical-Student-Education/Neurology-Clerkship/History/Sample-Write-Ups

Bates, 12th Edition Chapter 17

Summary

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- Importance of taking a good history when evaluating a neurological problem
- A neurological diagnosis can be obtained through performing a thorough examination that includes testing the motor and sensory systems, coordination, gait and reflexes in addition to the cranial nerves.
- Importance to know how to document the neurological examination.

References

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- Neurologic Differential Diagnosis- A Case - Based Approach, Alan B. Ettinger, MD,MBA and Deborah M. Weisbrot, MD- 2014; Chapter 1 (Introduction)

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- Textbook of Physical Diagnosis History and Examination, 7th Edition, Mark H. Swartz, MD, FACP, Chapter 17 (The Musculoskeletal System), Chapter 18 (The Nervous System), Chapter 19 (Putting the Examination Together)
- Textbook of Physical Diagnosis History and Examination, 8th Edition, Mark H. Swartz, MD, FACP, Chapter 21 (The Nervous System)

Preparation for Neuro Lab

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- Please be dressed in DPR lab attire
- Please bring a charged Ophthalmoscope, Visual Acuity card (Rosenbaum card), Penlight, Tuning Forks and Reflex Hammer
- Review Neurological DPR Skills Sheet as listed on the CPG

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Thank you

Lecture Feedback Form:

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